



NUTECH

National University of Technology
Department of Computer Engineering

Assignment 1

Course Title	: Discrete Structures	Course code	: MATH3901
Batch Number	: CEN-2024	Date	: 20-03-2025
Total Marks	: 10	Time Allowed	: 24-03-2025
FM Name	: Dr. Muhammad Sajjad	Semester	: Spring 2025

Question 1

[CLO-1/C3, PLO-1]

Apply the definitions of Equivalence Relation, Partially Ordered Set (POSET), and Function to determine whether the given relations R_1, R_2, R_3, R_4 and R_5 on A satisfy these properties. The relations are defined as follows:

- $R_1 = A \times A$.
- $R_2 = \{(+, +), (-, -), (\times, \times), (\div, \div)\}$.
- $R_3 = \{(+, +), (+, -), (-, -), (\times, \times), (\div, \div)\}$.
- $R_4 = \{(+, +), (+, -), (-, -), (-, +), (\times, \times), (\div, \div)\}$.
- $R_5 = \{(+, +), (-, -), (+, -), (\times, \times), (-, \times), (\div, \div)\}$.

Determine whether each relation is:

1. An Equivalence Relation (i.e., reflexive, symmetric, and transitive).
2. A Partially Ordered Set (POSET) (i.e., reflexive, antisymmetric, and transitive).
3. A Function.

Question 2

[CLO-1/C3, PLO-1]

Use the Venn Diagram concepts, draw the Venn Diagrams for the following cases:

- Given that $A \subseteq B$, $B \cap C \neq \emptyset$, and $A \cap C = \emptyset$, verify the Distributive Law.
- If A, B , and C are overlapping sets, verify the Distributive Law.
- If A, B , and C are disjoint sets, verify the Distributive Law.
- If A and B are disjoint, A and C are disjoint, but B and C are overlap, draw the Venn Diagram for the expression:

$$(A^c \cap (B \cup C)^c)^c.$$